

Brutal Practice Paper B41 - Forests, Respiration in Organisms, The Triangle & its Properties, Rational Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. In a forest food chain grass to deer to tiger, only 10% of energy passes to the next level. If grass holds 8000 units, how much energy finally reaches the tiger? [\[Forests\]](#)

- (A) 160 units; the deer doubles it
- (B) 800 units; energy decreases at each higher level
- (C) 8000 units; no energy is lost
- (D) 80 units; energy increases at each level
- (E) 8 units; energy decreases at each level
- (F) 720 units; energy is shared equally
- (G) 800 units; the tiger gets the most
- (H) 80 units; energy decreases at each higher level

2. An adult breathes about 16 times every minute. How many breaths does she take during 45 minutes of complete rest? [\[Respiration in Organisms\]](#)

- (A) 61 breaths
- (B) 360 breaths
- (C) 640 breaths
- (D) 450 breaths
- (E) 720 breaths
- (F) 7200 breaths

3. Two angles of a triangle are 48 deg and 67 deg. Find the exterior angle formed at the third vertex. [\[The Triangle & its Properties\]](#)

[Triangle & its Properties\]](#)

- (A) 115 deg
- (B) 130 deg
- (C) 67 deg
- (D) 95 deg
- (E) 65 deg
- (F) 48 deg
- (G) 113 deg
- (H) 125 deg

4. Add $-\frac{3}{4}$ and $\frac{5}{6}$, then multiply that sum by -6 . What is the final value? [\[Rational Numbers\]](#)

- (A) -6
- (B) $-\frac{7}{2}$
- (C) $\frac{1}{2}$
- (D) $-\frac{6}{12}$ left as is
- (E) $-\frac{1}{2}$
- (F) 6

5. A 200-hectare forest loses 15% of its area to clearing, then one-fourth of what remains is replanted. How many hectares are replanted? [\[Forests\]](#)
- (A) 17 hectares
 - (B) 30 hectares
 - (C) 42.5 hectares
 - (D) 85 hectares
 - (E) 50 hectares
 - (F) 127.5 hectares
6. At rest a person breathes 15 times a minute, but while running this rises to 35 times a minute. How many EXTRA breaths are taken in 3 minutes of running compared with resting? [\[Respiration in Organisms\]](#)
- (A) 150 extra breaths
 - (B) 105 extra breaths
 - (C) 45 extra breaths
 - (D) 75 extra breaths
 - (E) 60 extra breaths
 - (F) 30 extra breaths
7. A triangular signboard has its three corner angles in the ratio 2 : 3 : 4. Work out the largest of these angles. [\[The Triangle & its Properties\]](#)
- (A) 40 deg
 - (B) 20 deg
 - (C) 100 deg
 - (D) 72 deg
 - (E) 80 deg
 - (F) 90 deg
8. Divide $-\frac{8}{15}$ by $\frac{4}{5}$, then add $\frac{1}{3}$ to the quotient. What do you get? [\[Rational Numbers\]](#)
- (A) -1
 - (B) $-\frac{7}{15}$
 - (C) $-\frac{2}{3}$
 - (D) $\frac{1}{3}$
 - (E) $-\frac{11}{15}$
 - (F) $-\frac{1}{3}$
9. A forest has 5 tigers; each tiger needs 60 deer a year, and each deer needs 0.2 hectare of grassland. How many hectares of grassland feed the deer the tigers eat? [\[Forests\]](#)
- (A) 600 hectares
 - (B) 1500 hectares
 - (C) 60 hectares
 - (D) 65 hectares
 - (E) 120 hectares
 - (F) 300 hectares
 - (G) 30 hectares

10. Glucose is broken down to release energy. If one glucose unit releases 38 energy units, how much energy comes from 2.5 glucose units? [\[Respiration in Organisms\]](#)

- (A) 40.5 energy units
- (B) 90 energy units
- (C) 152 energy units
- (D) 19 energy units
- (E) 95.5 energy units
- (F) 76 energy units
- (G) 95 energy units

11. An exterior angle of a triangle is 124 deg and one of its interior opposite angles is 57 deg. Find the other interior opposite angle. [\[The Triangle & its Properties\]](#)

- (A) 57 deg
- (B) 67 deg
- (C) 47 deg
- (D) 113 deg
- (E) 23 deg
- (F) 33 deg

12. Which rational number lies exactly halfway between $-\frac{2}{3}$ and $\frac{1}{4}$? [\[Rational Numbers\]](#)

- (A) $-\frac{5}{12}$
- (B) $-\frac{5}{24}$
- (C) $\frac{5}{24}$
- (D) $-\frac{1}{2}$
- (E) $\frac{5}{12}$
- (F) $-\frac{7}{24}$
- (G) $-\frac{5}{7}$

13. A single tree absorbs about 22 kg of carbon dioxide per year. A small forest of 350 such trees absorbs how much carbon dioxide in one year? [\[Forests\]](#)

- (A) 7000 kg
- (B) 77000 kg
- (C) 700 kg
- (D) 770 kg
- (E) 7700 kg
- (F) 15400 kg
- (G) 3850 kg
- (H) 372 kg

14. A person inhales about 500 mL of air per breath and breathes 18 times a minute. How many litres of air are breathed in 5 minutes? [\[Respiration in Organisms\]](#)

- (A) 4.5 litres
- (B) 45 millilitres
- (C) 9 litres
- (D) 45 litres
- (E) 90 litres
- (F) 40 litres

- 15.** One base angle of an isosceles triangle is 52 deg. Find its vertex angle, given the two base angles are equal. [\[The Triangle & its Properties\]](#)
- (A) 128 deg
 - (B) 88 deg
 - (C) 76 deg
 - (D) 26 deg
 - (E) 64 deg
 - (F) 52 deg
 - (G) 104 deg
- 16.** Write $-36/48$ in standard form (lowest terms with a positive denominator). [\[Rational Numbers\]](#)
- (A) $3/4$
 - (B) $-36/48$
 - (C) $36/-48$
 - (D) $-9/12$
 - (E) $-3/4$
 - (F) $-18/24$
- 17.** In a forest patch the ratio of producers to herbivores to carnivores is 50 : 10 : 1. If there are 2 carnivores, how many producers are there? [\[Forests\]](#)
- (A) 200 producers
 - (B) 10 producers
 - (C) 50 producers
 - (D) 500 producers
 - (E) 61 producers
 - (F) 100 producers
 - (G) 25 producers
- 18.** A resting baby breathes 12 times a minute. A doctor counts the breaths over just 20 seconds. How many breaths should she expect in those 20 seconds? [\[Respiration in Organisms\]](#)
- (A) 3 breaths
 - (B) 20 breaths
 - (C) 8 breaths
 - (D) 12 breaths
 - (E) 240 breaths
 - (F) 4 breaths
 - (G) 6 breaths
 - (H) 2 breaths
- 19.** The two equal sides of an isosceles triangle make a vertex angle of 40 deg. Find each base angle. [\[The Triangle & its Properties\]](#)
- (A) 110 deg
 - (B) 70 deg
 - (C) 140 deg
 - (D) 80 deg
 - (E) 50 deg
 - (F) 35 deg

20. Subtract $-\frac{5}{6}$ from $-\frac{1}{3}$. What is the result? [\[Rational Numbers\]](#)

- (A) 1
- (B) $\frac{1}{2}$
- (C) $\frac{1}{6}$
- (D) $-\frac{2}{3}$
- (E) $-\frac{7}{6}$
- (F) $\frac{7}{6}$
- (G) $-\frac{1}{6}$

21. Decomposition makes a 6 kg pile of leaf litter lose two-thirds of its mass as it turns into humus. What mass of humus-rich material remains? [\[Forests\]](#)

- (A) 4 kg
- (B) 1.5 kg
- (C) 2 kg
- (D) 0.5 kg
- (E) 3 kg
- (F) 1 kg

22. A muscle cell uses 6 mL of oxygen each minute and works steadily for 7.5 minutes. How much oxygen does it use in total? [\[Respiration in Organisms\]](#)

- (A) 45 litres
- (B) 48 mL
- (C) 42 mL
- (D) 13.5 mL
- (E) 0.8 mL
- (F) 450 mL
- (G) 45 mL
- (H) 36 mL

23. A ramp shaped like a right-angled triangle stands on two perpendicular legs measuring 9 cm and 12 cm. How long is its sloping hypotenuse? [\[The Triangle & its Properties\]](#)

- (A) 21 cm
- (B) 18 cm
- (C) 11 cm
- (D) 13 cm
- (E) 7.5 cm
- (F) 15 cm
- (G) 20 cm
- (H) 225 cm

24. The product of two rational numbers is $-\frac{16}{9}$. If one of them is $-\frac{4}{3}$, find the other number. [\[Rational Numbers\]](#)

- (A) $-\frac{4}{3}$
- (B) $\frac{4}{9}$
- (C) $\frac{12}{9}$
- (D) $-\frac{12}{9}$
- (E) $\frac{4}{3}$
- (F) $\frac{3}{4}$

25. A forest department plants 24 saplings per row in 15 rows, and only five-sixths of them survive. How many saplings survive? [\[Forests\]](#)

- (A) 250 saplings
- (B) 270 saplings
- (C) 360 saplings
- (D) 300 saplings
- (E) 288 saplings
- (F) 330 saplings

26. A runner's breathing rate jumps from 16 to 40 breaths per minute. By what percentage did the breathing rate increase? [\[Respiration in Organisms\]](#)

- (A) 100%
- (B) 150%
- (C) 15%
- (D) 2.5%
- (E) 40%
- (F) 250%
- (G) 24%
- (H) 60%

27. A right-angled triangle has a hypotenuse of 25 cm and one leg of 7 cm. Find the length of the other leg.

[\[The Triangle & its Properties\]](#)

- (A) 26 cm
- (B) 23 cm
- (C) 20 cm
- (D) 32 cm
- (E) 18 cm
- (F) 24 cm
- (G) 576 cm
- (H) 12.5 cm

28. By what rational number should $-\frac{8}{9}$ be multiplied to give $\frac{2}{3}$? [\[Rational Numbers\]](#)

- (A) $\frac{3}{4}$
- (B) $-\frac{3}{8}$
- (C) $\frac{16}{27}$
- (D) $-\frac{3}{4}$
- (E) $-\frac{16}{27}$
- (F) $-\frac{4}{3}$
- (G) $\frac{4}{3}$

29. If a forest canopy intercepts 80% of the sunlight, what fraction reaches the floor, and what does that mean for the plants there? [\[Forests\]](#)

- (A) Four-fifths (80%) reaches the floor, so many plants grow
- (B) One-fifth (20%) reaches, so no plant can ever grow
- (C) One-fifth (20%) reaches, so tall trees grow there
- (D) All the sunlight reaches the floor
- (E) Half (50%) reaches the floor
- (F) Three-fifths (60%) reaches the floor
- (G) One-fifth (20%) reaches the floor, so only shade-loving plants grow there
- (H) None reaches, so the floor is bare rock

30. A hospital oxygen cylinder holds 3000 litres. A patient uses 0.5 litre per breath and breathes 20 times a minute. For about how many minutes will the cylinder last? [\[Respiration in Organisms\]](#)

- (A) 150 minutes
- (B) 1500 minutes
- (C) 300 minutes
- (D) 200 minutes
- (E) 3000 minutes
- (F) 30 minutes
- (G) 600 minutes

31. Can a triangle be formed with sides 5 cm, 6 cm and 12 cm? State the correct reason. [\[The Triangle & its Properties\]](#)

- (A) Yes, because all three sides are positive
- (B) Yes, because the three sides are all different
- (C) Yes, it forms a right-angled triangle
- (D) No, because 12 is an even number
- (E) No, because $5 + 12$ is not greater than 6
- (F) Yes, because 12 is the longest side
- (G) No, because $6 + 12$ is less than 5
- (H) No, because $5 + 6$ is not greater than 12

32. Among $-\frac{3}{4}$, $-\frac{2}{3}$ and $-\frac{5}{6}$, which is the smallest? [\[Rational Numbers\]](#)

- (A) $-\frac{1}{2}$
- (B) $-\frac{5}{6}$
- (C) $\frac{5}{6}$
- (D) $-\frac{2}{3}$
- (E) $-\frac{3}{4}$
- (F) $-\frac{7}{12}$

33. A forest produces about 30 kg of oxygen per tree per year, and a family of four needs roughly 9000 kg of oxygen a year. About how many such trees supply this family? [\[Forests\]](#)

- (A) About 300 trees
- (B) About 36 trees
- (C) About 30 trees
- (D) About 150 trees
- (E) About 3000 trees
- (F) About 900 trees

- 34.** During heavy exercise our muscles may respire without enough oxygen. What substance builds up and causes muscle cramps? [\[Respiration in Organisms\]](#)
- (A) Vinegar
 - (B) Only water
 - (C) Alcohol and carbon dioxide
 - (D) Pure starch
 - (E) Carbon monoxide
 - (F) Lactic acid
- 35.** A triangle has sides 8 cm, 15 cm and 17 cm. Is it right-angled, and why? [\[The Triangle & its Properties\]](#)
- (A) Yes, because $15 - 8 = 7$
 - (B) Yes, because all three sides are different
 - (C) Yes, because $8 + 15 = 23$
 - (D) Yes, because $8^2 + 15^2 = 17^2$
 - (E) No, because none of the sides are equal
 - (F) No, because 17 is a prime number
- 36.** Find the additive inverse of $(-\frac{3}{5} + \frac{7}{10})$. [\[Rational Numbers\]](#)
- (A) $\frac{13}{10}$
 - (B) $\frac{1}{10}$
 - (C) $-\frac{13}{10}$
 - (D) $-\frac{1}{10}$
 - (E) 10
 - (F) $\frac{1}{5}$
- 37.** Which forest layer receives the most sunlight and is formed by the leafy tops of the tallest trees? [\[Forests\]](#)
- (A) The root zone underground
 - (B) The understorey, made of shrubs
 - (C) The canopy, the green roof formed by tall tree tops
 - (D) The herb layer near the ground
 - (E) The litter of dead leaves
 - (F) The forest floor, where seeds sprout
 - (G) The decomposer layer
 - (H) The soil layer rich in humus
- 38.** Yeast respire without oxygen when bread dough rises. This anaerobic respiration produces: [\[Respiration in Organisms\]](#)
- (A) Alcohol and carbon dioxide
 - (B) Only water vapour
 - (C) Oxygen and water
 - (D) Lactic acid only
 - (E) Glucose and oxygen
 - (F) Carbon monoxide
 - (G) Vinegar and salt

39. Two sides of a triangle are 7 cm and 10 cm. Between what limits must the length of the third side lie?

[The Triangle & its Properties]

- (A) Between 0 cm and 3 cm
- (B) Greater than 10 cm and less than 17 cm
- (C) Between 7 cm and 10 cm
- (D) Exactly 3 cm
- (E) Greater than 3 cm and less than 17 cm
- (F) Between 3 cm and 7 cm
- (G) Exactly 17 cm

40. Find the reciprocal of $(-\frac{2}{7} - \frac{3}{14})$. [Rational Numbers]

- (A) 2
- (B) $-\frac{1}{2}$
- (C) -7
- (D) -14
- (E) $\frac{1}{2}$
- (F) -2

41. Dead leaves and animal remains are turned into dark, nutrient-rich humus chiefly by which living things? [Forests]

- (A) The heat coming from inside the Earth
- (B) Sunlight acting alone
- (C) Birds nesting in the trees
- (D) Flowing rainwater only
- (E) Microorganisms such as fungi and bacteria
- (F) Large herbivores like deer
- (G) Tigers and other carnivores
- (H) Green plants during photosynthesis

42. When we blow into clear lime water it turns milky. This proves that the air we breathe out is rich in:

[Respiration in Organisms]

- (A) Alcohol
- (B) Carbon dioxide
- (C) Lactic acid
- (D) Nitrogen
- (E) Oxygen
- (F) Water vapour
- (G) Helium

43. In triangle ABC, angle A = 90 deg and angle B = 35 deg. Find the exterior angle at vertex C. [The Triangle & its Properties]

[The Triangle & its Properties]

- (A) 120 deg
- (B) 125 deg
- (C) 35 deg
- (D) 55 deg
- (E) 115 deg
- (F) 90 deg
- (G) 145 deg
- (H) 70 deg

44. Simplify: $(-3/4)$ multiplied by $(8/9)$, then divided by $(-2/3)$. [\[Rational Numbers\]](#)

- (A) $-3/2$
- (B) $-2/3$
- (C) $2/3$
- (D) $-4/9$
- (E) $4/9$
- (F) $3/2$
- (G) 1

45. Why is a forest often called the 'green lungs' of a region? [\[Forests\]](#)

- (A) Trees give out carbon dioxide for animals to use
- (B) Trees breathe in oxygen only at night
- (C) Forests make the air completely still
- (D) The leaves are green just like lungs
- (E) Forests only store water like a sponge
- (F) Forests block all the sunlight
- (G) Trees release oxygen and absorb carbon dioxide, keeping the air balanced

46. Fish take in the oxygen dissolved in water using which special organs? [\[Respiration in Organisms\]](#)

- (A) Gills
- (B) Eyes
- (C) Tail fins
- (D) Stomata
- (E) Skin only
- (F) Scales
- (G) Lungs
- (H) Nostrils

47. A median of a triangle joins a vertex to the midpoint of the opposite side. How do the areas of the two triangles it creates compare? [\[The Triangle & its Properties\]](#)

- (A) The areas are in the ratio 1 : 4
- (B) They have equal areas
- (C) They differ by the length of the base
- (D) It cannot be determined
- (E) They are in the ratio 2 : 1
- (F) The larger half is double the smaller

48. Counting carefully, how many different rational numbers can be found lying strictly between $-1/2$ and $1/2$ on the number line? [\[Rational Numbers\]](#)

- (A) Exactly 1
- (B) Infinitely many
- (C) Only 0
- (D) Exactly 9
- (E) Exactly 2
- (F) Zero
- (G) Exactly 10
- (H) Exactly 5

49. What would most likely happen to a forest if all its decomposers suddenly vanished? [\[Forests\]](#)

- (A) The soil would instantly turn to rock
- (B) All animals would become herbivores
- (C) The forest would double in size
- (D) Dead plants and animals would pile up and nutrients would not return to the soil
- (E) Rainfall would immediately stop
- (F) Nothing at all would change
- (G) Oxygen would increase sharply
- (H) The trees would grow twice as fast

50. Air enters an insect's body for respiration through tiny openings that lead to fine tubes. These openings are the: [\[Respiration in Organisms\]](#)

- (A) The mouth leading to lungs
- (B) Hairs covering the body
- (C) The skin over the whole body
- (D) A single pair of lungs
- (E) Gills on the sides of the body
- (F) The wings
- (G) The antennae
- (H) Spiracles leading to tracheae

51. An equilateral triangle has a perimeter of 54 cm. What is the value of one interior angle (in degrees) added to one side length (in cm)? [\[The Triangle & its Properties\]](#)

- (A) 78
- (B) 72
- (C) 60
- (D) 108
- (E) 114
- (F) 90

52. The sum of a rational number and $-\frac{7}{8}$ is $\frac{1}{4}$. Find the rational number. [\[Rational Numbers\]](#)

- (A) $-\frac{5}{4}$
- (B) $\frac{9}{4}$
- (C) $-\frac{1}{8}$
- (D) $\frac{9}{8}$
- (E) $-\frac{5}{8}$
- (F) $-\frac{9}{8}$
- (G) $\frac{1}{8}$
- (H) $\frac{5}{8}$

53. How do forests help reduce the risk of floods after heavy rain? [\[Forests\]](#)

- (A) Roots hold the soil and let rainwater seep slowly into the ground
- (B) Animals dig channels for the water
- (C) Forests heat the air to dry the rain
- (D) Trees release water only to cause rain
- (E) Trees turn rainwater into oxygen
- (F) Leaves push the rain clouds away
- (G) Trees quickly drink up all the floodwater
- (H) Bark blocks rivers from flowing

54. Earthworms, which have no lungs or gills, breathe through their: [\[Respiration in Organisms\]](#)

- (A) Mouth only
- (B) Tail
- (C) Body hair
- (D) Lungs
- (E) Spiracles
- (F) Moist skin
- (G) Stomata
- (H) Gills

55. In an isosceles triangle the vertex angle is twice each base angle. Find the vertex angle. [\[The Triangle & its Properties\]](#)

- (A) 80 deg
- (B) 120 deg
- (C) 45 deg
- (D) 36 deg
- (E) 60 deg
- (F) 90 deg

56. Evaluate $-\frac{5}{12} + \frac{7}{18}$ using the LCM of 12 and 18. [\[Rational Numbers\]](#)

- (A) $\frac{29}{36}$
- (B) $\frac{1}{3}$
- (C) $\frac{2}{30}$
- (D) $-\frac{1}{36}$
- (E) $-\frac{1}{6}$
- (F) $\frac{1}{36}$
- (G) $-\frac{29}{36}$

57. What exactly is humus in a forest soil? [\[Forests\]](#)

- (A) A flowering forest shrub
- (B) Rainwater stored inside roots
- (C) The green roof of the forest
- (D) A type of tall forest tree
- (E) Dark material formed from rotted dead plants and animals that enriches the soil
- (F) A young growing sapling
- (G) A gas that trees breathe out
- (H) Hard rock found under the soil

58. Plant leaves take in oxygen and release carbon dioxide during respiration mainly through tiny pores called: [\[Respiration in Organisms\]](#)

- (A) The chlorophyll
- (B) The waxy cuticle
- (C) The flower petals
- (D) Stomata in the leaves
- (E) The veins in the leaves
- (F) The phloem tubes

59. If you take one exterior angle at each of the three corners of a triangle, what do all three add up to? [\[The Triangle & its Properties\]](#)

[Triangle & its Properties\]](#)

- (A) 450 deg
- (B) 270 deg
- (C) 720 deg
- (D) 540 deg
- (E) 300 deg
- (F) 90 deg
- (G) 180 deg
- (H) 360 deg

60. The gap between $-\frac{2}{5}$ and $\frac{4}{5}$ on a number line is split into 3 equal parts. What is the length of each part? [\[Rational Numbers\]](#)

- (A) $\frac{3}{5}$
- (B) $\frac{6}{5}$
- (C) $\frac{2}{15}$
- (D) $\frac{1}{5}$
- (E) $\frac{6}{15}$
- (F) $\frac{2}{5}$
- (G) $\frac{2}{3}$

61. Which sequence correctly shows how energy flows through a forest? [\[Forests\]](#)

- (A) Sun to green plants to herbivores to carnivores
- (B) Sun to herbivores to plants to carnivores
- (C) Sun to carnivores to herbivores to plants
- (D) Green plants to Sun to carnivores
- (E) Soil to Sun to plants to animals
- (F) Plants to soil to Sun to animals
- (G) Herbivores to plants to Sun
- (H) Carnivores to herbivores to plants to Sun

62. Why does an athlete pant - breathing fast and deep - for a while after finishing a sprint? [\[Respiration in Organisms\]](#)

[Respiration in](#)

- (A) To take in extra carbon dioxide
- (B) To take in more oxygen and break down the lactic acid built up in the muscles
- (C) To remove water from the blood quickly
- (D) To make even more lactic acid fast
- (E) To release the extra oxygen the muscles made
- (F) To warm up the leg muscles

63. A right-angled isosceles triangle has two equal legs of 6 cm each. Find its hypotenuse (use square root of 2 = 1.41). [\[The Triangle & its Properties\]](#)

- (A) About 12 cm
- (B) About 36 cm
- (C) About 4.24 cm
- (D) About 7.41 cm
- (E) About 8.46 cm
- (F) About 9 cm

64. Which is greater, $-\frac{5}{7}$ or $-\frac{7}{9}$, and by how much? [\[Rational Numbers\]](#)
- (A) $-\frac{5}{7}$ is greater, by $\frac{12}{63}$
 - (B) $-\frac{5}{7}$ is greater, by $\frac{1}{63}$
 - (C) They are equal
 - (D) $-\frac{7}{9}$ is greater, by $\frac{2}{63}$
 - (E) $-\frac{7}{9}$ is greater, by $\frac{12}{63}$
 - (F) $-\frac{5}{7}$ is greater, by $\frac{4}{63}$
65. What is the main role of herbivores in a forest, linking plants to carnivores? [\[Forests\]](#)
- (A) They make their own food from sunlight
 - (B) They hunt and eat only other carnivores
 - (C) They eat plants and become food for carnivores, passing energy upward
 - (D) They live only deep underground
 - (E) They produce oxygen for the whole forest
 - (F) They turn soil into solid rock
 - (G) They decompose dead leaves into humus
66. The large muscular sheet below the lungs that helps in breathing is the: [\[Respiration in Organisms\]](#)
- (A) Diaphragm
 - (B) Ribcage
 - (C) Heart
 - (D) Liver
 - (E) Gills
 - (F) Windpipe
 - (G) Backbone
 - (H) Stomach
67. The three angles of a triangle are $2x$, $3x$ and $(4x + 9)$ degrees. Find the value of x . [\[The Triangle & its Properties\]](#)
- (A) 18
 - (B) 171
 - (C) 17
 - (D) 19
 - (E) 21
 - (F) 9
 - (G) 20
68. Multiply $-\frac{2}{3}$ by its own additive inverse. What is the product? [\[Rational Numbers\]](#)
- (A) $-\frac{4}{3}$
 - (B) 1
 - (C) -1
 - (D) $\frac{2}{3}$
 - (E) $-\frac{2}{3}$
 - (F) $\frac{4}{9}$
 - (G) $-\frac{4}{9}$
 - (H) 0

69. Why does cutting down large areas of forest increase the carbon dioxide in the air? [\[Forests\]](#)

- (A) Cutting trees has no effect on the air
- (B) The cut trees release stored oxygen
- (C) Animals breathe faster without shade
- (D) Carbon dioxide is heavier without trees
- (E) The Sun gets hotter without trees
- (F) Fewer trees remain to absorb carbon dioxide during photosynthesis
- (G) Rain washes carbon dioxide into the air

70. During inhalation, what do the diaphragm and ribs do? [\[Respiration in Organisms\]](#)

- (A) The ribs and diaphragm both move downward together
- (B) Only the stomach moves while breathing in
- (C) The diaphragm rises and the ribs move in, shrinking the chest
- (D) Both the diaphragm and ribs stay completely still
- (E) The diaphragm flattens (moves down) and the ribs move up and out, expanding the chest
- (F) The diaphragm flattens but the ribs move inward

71. Two angles of a triangle measure $3x$ and $2x$, and the third angle is 80° . Find the larger of the two unknown angles. [\[The Triangle & its Properties\]](#)

- (A) 30°
- (B) 50°
- (C) 72°
- (D) 48°
- (E) 100°
- (F) 40°
- (G) 60°

72. Multiply $-\frac{5}{6}$ by its own reciprocal. What is the product? [\[Rational Numbers\]](#)

- (A) -1
- (B) $\frac{11}{30}$
- (C) $-\frac{6}{5}$
- (D) $-\frac{5}{6}$
- (E) 1
- (F) $\frac{25}{36}$
- (G) $-\frac{25}{36}$

73. Why is the forest floor relatively dark even on a bright sunny day? [\[Forests\]](#)

- (A) The trees switch off light at the top
- (B) There are no leaves anywhere above
- (C) Animals cover the whole ground
- (D) The soil gives off black smoke
- (E) The dense canopy of leaves above blocks most of the sunlight
- (F) The floor absorbs all colours of light

74. Anaerobic respiration releases how much energy compared with aerobic respiration from the same glucose? [\[Respiration in Organisms\]](#)

- (A) Slightly more energy than aerobic
- (B) Energy only in the form of heat
- (C) An unlimited amount of energy
- (D) Less energy, because the glucose is only partly broken down
- (E) More energy, because no oxygen is needed
- (F) Exactly the same energy as aerobic
- (G) No energy at all
- (H) Twice as much energy

75. The altitude of a triangle from a vertex is best described as the segment that is: [\[The Triangle & its Properties\]](#)

- (A) A line joining two midpoints
- (B) Always equal in length to the median
- (C) A line drawn parallel to the base
- (D) The longest side of the triangle
- (E) Always half the longest side
- (F) A line that bisects the angle at the vertex
- (G) Perpendicular from the vertex to the opposite side

76. Divide the sum of $-\frac{1}{2}$ and $-\frac{1}{3}$ by the product of $-\frac{1}{2}$ and $-\frac{1}{3}$. [\[Rational Numbers\]](#)

- (A) $-\frac{1}{6}$
- (B) -5
- (C) $\frac{1}{5}$
- (D) $-\frac{1}{5}$
- (E) -30
- (F) 5
- (G) $-\frac{5}{36}$
- (H) $-\frac{5}{6}$

77. Animal dung and dead remains are useful in a forest mainly because they: [\[Forests\]](#)

- (A) Are eaten directly by green plants
- (B) Turn straight into new trees overnight
- (C) Stop seeds from sprouting
- (D) Have no effect on the forest at all
- (E) Poison the groundwater permanently
- (F) Decompose and return nutrients to the soil for plants
- (G) Make the soil unable to hold water
- (H) Block sunlight from reaching the trees

78. Which organ pair takes in oxygen for the whole human body? [\[Respiration in Organisms\]](#)

- (A) The lungs
- (B) The stomach
- (C) The gills
- (D) The skin pores
- (E) The brain
- (F) The heart
- (G) The liver

79. In triangle PQR, the exterior angle at Q is 105 deg, and angle P is twice angle R. Find angle R. [\[The](#)

[Triangle & its Properties\]](#)

- (A) 45 deg
- (B) 30 deg
- (C) 35 deg
- (D) 52.5 deg
- (E) 60 deg
- (F) 70 deg
- (G) 40 deg

80. Three-fourths of a rational number is $-\frac{9}{10}$. Find the number. [\[Rational Numbers\]](#)

- (A) $\frac{27}{40}$
- (B) $-\frac{27}{40}$
- (C) $-\frac{12}{5}$
- (D) $-\frac{3}{5}$
- (E) $-\frac{6}{5}$
- (F) $\frac{3}{5}$
- (G) $\frac{6}{5}$
- (H) $-\frac{9}{10}$

81. Saplings in a forest are best described as: [\[Forests\]](#)

- (A) Fungi on the forest floor
- (B) A kind of forest gas
- (C) Fully grown old trees
- (D) Young plants that grow into trees
- (E) Underground tree roots
- (F) Animals that eat seeds
- (G) The topmost branches of a tree
- (H) Dead and decaying leaves

82. Breathing is only one part of respiration. Where does the actual release of energy from food take place? [\[Respiration in Organisms\]](#)

- (A) In the throat
- (B) In the stomach
- (C) In the blood vessels only
- (D) In the nose
- (E) In the diaphragm
- (F) In the cells of the body

83. A right-angled triangle has legs of 1.5 m and 2 m. Find its hypotenuse in metres. [\[The Triangle & its Properties\]](#)

- (A) 2.25 m
- (B) 2.5 m
- (C) 1.75 m
- (D) 2 m
- (E) 4 m
- (F) 5 m
- (G) 3.5 m

84. Evaluate $(-2/5)$ squared, then divide the result by $(-4/15)$. [\[Rational Numbers\]](#)

- (A) $4/25$
- (B) $-5/3$
- (C) $3/5$
- (D) $-4/25$
- (E) $-3/5$
- (F) $5/3$

85. How does a forest act like a natural water filter and help recharge groundwater? [\[Forests\]](#)

- (A) Trees boil the water clean with heat
- (B) Leaves purify water using strong acid
- (C) Bark filters the water like a cloth sieve
- (D) Rainwater passes slowly through leaves and soil, seeping down to refill groundwater
- (E) Trees freeze the water to clean it
- (F) Forests have no effect on groundwater

86. Why can't humans breathe normally underwater the way fish do? [\[Respiration in Organisms\]](#)

- (A) Water is too heavy to breathe
- (B) Our lungs cannot take the oxygen dissolved in water; they need air
- (C) We can breathe out but never in underwater
- (D) Fish make oxygen for us to use
- (E) Water turns oxygen into nitrogen
- (F) Our gills stop working in water
- (G) Our lungs are simply too small for water

87. Each base angle of an isosceles triangle is $(x + 15)$ degrees and the vertex angle is x degrees. Find x .

[\[The Triangle & its Properties\]](#)

- (A) 65 deg
- (B) 55 deg
- (C) 40 deg
- (D) 60 deg
- (E) 80 deg
- (F) 45 deg
- (G) 70 deg
- (H) 50 deg

88. A debt is written as $-5/8$ of a unit and a credit as $3/4$ of a unit. What is the net balance? [\[Rational Numbers\]](#)

- (A) $-11/8$
- (B) $-1/8$
- (C) $1/2$
- (D) $11/8$
- (E) $-1/4$
- (F) $1/8$
- (G) $-1/2$

89. Why are forests important for keeping the oxygen supply in the air steady? [\[Forests\]](#)

- (A) Decomposers release most of the oxygen
- (B) Animals breathe out oxygen for the trees
- (C) The soil produces oxygen at night
- (D) Tree bark traps oxygen from the rain
- (E) Trees store oxygen in their roots taken from soil
- (F) During photosynthesis the trees release oxygen into the air

90. When you hold your breath, carbon dioxide builds up in your blood. How does the body usually respond? [\[Respiration in Organisms\]](#)

- (A) It cools the blood to slow breathing
- (B) It produces fresh oxygen inside the blood
- (C) It stops the heart to save oxygen
- (D) It forces a strong urge to breathe so the carbon dioxide can be removed
- (E) It does nothing at all
- (F) It stores the carbon dioxide in the muscles

91. A 13 m ladder leans against a wall with its foot 5 m away from the wall. How high up the wall does the ladder reach? [\[The Triangle & its Properties\]](#)

- (A) 144 m
- (B) 8 m
- (C) 9 m
- (D) 12 m
- (E) 18 m
- (F) 12.5 m
- (G) 11 m

92. Express $-\frac{7}{3}$ as a mixed number. [\[Rational Numbers\]](#)

- (A) $-3\frac{2}{3}$
- (B) $-2\frac{1}{3}$
- (C) $-3\frac{1}{3}$
- (D) $-2\frac{1}{2}$
- (E) $-2\frac{2}{3}$
- (F) $-1\frac{1}{3}$
- (G) $2\frac{1}{3}$

93. The continuous green roof of a forest, formed by the overlapping upper branches of many trees, is called the: [\[Forests\]](#)

- (A) Sapling
- (B) Leaf litter
- (C) Canopy
- (D) Humus
- (E) Understorey
- (F) Forest floor

94. Which gases are mainly taken in and given out during normal human breathing? [\[Respiration in Organisms\]](#)

- (A) Water vapour is taken in; oxygen is given out
- (B) Hydrogen is taken in; carbon dioxide is given out
- (C) Oxygen is both taken in and given out
- (D) Oxygen is taken in; carbon dioxide is given out
- (E) Carbon dioxide is taken in; oxygen is given out
- (F) Carbon dioxide is taken in; nitrogen is given out

95. In a triangle one angle is a right angle and the other two are equal. Find each of the two equal angles.

[\[The Triangle & its Properties\]](#)

- (A) 55 deg
- (B) 60 deg
- (C) 90 deg
- (D) 30 deg
- (E) 35 deg
- (F) 75 deg
- (G) 45 deg
- (H) 50 deg

96. If $x = -1/2$ and $y = 2/3$, evaluate $x - y + xy$. [\[Rational Numbers\]](#)

- (A) $-1/3$
- (B) $-1/6$
- (C) $3/2$
- (D) $-3/2$
- (E) $5/6$
- (F) $-11/6$
- (G) $-5/6$
- (H) $-7/6$

97. Which statement best explains why a forest is called a 'dynamic living entity'? [\[Forests\]](#)

- (A) It stays completely frozen all year
- (B) It is made only of non-living rocks and water
- (C) It is built by humans like a building
- (D) Nothing inside it ever changes over time
- (E) It is empty of any living thing
- (F) It is full of life and keeps changing as plants and animals grow, die and recycle

98. On a cold day we can see a faint mist when we breathe out. This shows that the air we exhale contains: [\[Respiration in Organisms\]](#)

- (A) Pure oxygen
- (B) Water vapour
- (C) Lactic acid gas
- (D) Smoke
- (E) Carbon only
- (F) Nitrogen only

99. The angles of a triangle are $(x - 10)$, x and $(x + 40)$ degrees. Find the smallest angle. [\[The Triangle & its Properties\]](#)

- (A) 60 deg
- (B) 50 deg
- (C) 20 deg
- (D) 40 deg
- (E) 80 deg
- (F) 30 deg

100. The temperature changes by $-\frac{3}{4}$ degree each hour for 5 hours, then rises by 2 degrees. What is the net change? [\[Rational Numbers\]](#)

- (A) -2 degree
- (B) $\frac{15}{4}$ degree
- (C) $-\frac{15}{4}$ degree
- (D) $\frac{7}{4}$ degree
- (E) $-\frac{1}{4}$ degree
- (F) $-\frac{23}{4}$ degree
- (G) $-\frac{7}{4}$ degree